

Image editing and its advance applications : Inpainting / segmentation / Style Transfer / Deblurring / colorization

Deep learning : Computer Vision and Image processing with Machine Learning

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Abstract—Large text-to-image models have significantly advanced AI by generating high-quality, diverse images from text prompts. However, these models struggle to accurately replicate the appearance of specific subjects from reference images. By fine-tuning a pretrained model with a few reference images, this approach creates a unique identifier for the subject, enabling photorealistic generation of the subject in various scenes, poses, and lighting conditions. This technique employs a class-specific prior preservation loss, facilitating tasks such as subject recontextualization, text-guided view synthesis, and artistic rendering, while preserving key features of the subject. The paper also introduces a new dataset and evaluation protocol for subject-driven generation.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

Recent advancements in image synthesis have transformed the field of computer vision. High-resolution image generation, once computationally intensive, has become more efficient and high-quality due to methods like GANs and diffusion models. Text-to-image models now generate realistic images from text prompts using large-scale image-caption datasets. However, these models still face challenges in accurately replicating specific subjects, often producing variations of the target. To address this, multimodal learning and retrieval-augmented models have been developed, improving photorealism and the handling of rare entities. These innovations have broad applications, including inpainting, segmentation, style transfer, deblurring, and colorization.

II. LITERATURE SURVEY

Generative models exhibit varying strengths and limitations in image synthesis. GANs are efficient for generating high-resolution images but are difficult to optimize and often struggle to capture the full data distribution. VAEs and flow-based models, while easier to optimize and providing good density estimation, produce lower sample quality compared to GANs. Autoregressive models excel in density estimation

but are computationally intensive and limited to low-resolution output. Diffusion models (DMs) represent the state-of-the-art in both density estimation and sample quality, though they require high training costs and slow inference speeds. Latent Diffusion Models (LDMs), on the other hand, improve efficiency by operating in a compressed latent space, reducing training costs and speeding up inference while maintaining high quality. Two-stage approaches, such as VQ-VAEs and VQGANs, combine different models to scale autoregressive techniques, though they introduce trade-offs in compression and computational complexity. Among these models, LDMs stand out by balancing compression and generative quality, ensuring high-fidelity reconstructions with greater scalability compared to others.

III. LIMITATIONS

Following are the limitations based on the respective research papers

A. Dreambooth : Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation

The model faces limitations in generating accurate images for rare contexts due to weak priors or the low probability of co-occurrence between subjects and concepts in the training set. Context-appearance entanglement may occur, where the subject's appearance changes based on the context. The model can also overfit, producing images too similar to the training data, especially when prompts resemble the original environment. Some subjects, like dogs and cats, are easier to learn, while rarer subjects may show fewer variations. Additionally, subject fidelity can vary, with some images containing hallucinated features depending on the model's prior and semantic complexity.

B. Re-imagen retrieval Augmented

The quality of generated images in models like Re-Imagen is highly dependent on the accuracy and relevance of the retrieved image-text pairs, where low-

quality retrievals can degrade image fidelity. Additionally, Re-Imagen faces challenges when handling complex prompts involving multiple entities or highly compositional scenes. For instance, with a prompt like "A cat wearing a chef's hat and apron, cooking a meal in a tiny kitchen," the model may struggle to depict proportions and spatial relationships accurately, resulting in incoherent elements. Furthermore, the model's super-resolution capabilities sometimes fail to capture fine-grained details, leading to visual distortions or artifacts, particularly in high-resolution images. For example, a prompt like "A close-up of a butterfly with intricate patterns on its wings" may result in blurry or distorted wing patterns, possibly due to limitations in the super-resolution algorithm or insufficient training data on such intricate subjects.

C. High-Resolution Image Synthesis with Latent Diffusion Models

While Latent Diffusion Models (LDMs) greatly reduce the computational demands compared to methods that work directly with pixels, they are still slower than Generative Adversarial Networks (GANs) due to their step-by-step process. Additionally, LDMs might not be the best choice when very high precision is needed. Even though the loss in image quality is minimal in our compressed models their ability to reconstruct images can become a limitation for tasks that need extremely accurate details at the pixel level. We believe that our superresolution models may already be somewhat restricted in this area.

D. Photorealistic Text-to-Image Diffusion Models with Deep Language Understanding

Dataset Limitations: The training data for Imagen comprises large, uncensored, web-scraped datasets, which have been critiqued for containing harmful content, such as pornographic imagery, racist slurs, and social stereotypes. Despite efforts to filter undesirable content, such datasets may perpetuate and amplify biases and harmful associations, particularly towards marginalized groups. This raises ethical concerns about the model's potential for representational harm.

Biases in Text Encoders: Imagen utilizes text encoders trained on extensive web-scale data, inheriting inherent social biases and limitations of large language models. This includes biases in generating images, such as a tendency to produce images of individuals with lighter skin tones and alignment with Western gender stereotypes in professional contexts.

Image Quality and Fidelity: Imagen encounters challenges in generating high-fidelity images, particularly those depicting people. Internal evaluations indicate that the model performs better with non-human images, suggesting a degradation in image quality and accuracy when portraying human subjects.

Mode Collapse and Bias Amplification: Like other generative models, Imagen risks mode collapse, where certain modes of the data distribution are underrepresented or omitted, potentially exacerbating biases present in the training data.

Ethical and Responsible Use: Given the potential for misuse in generating harmful content or misinformation, Imagen's release is restricted. The ethical implications of such models highlight the need for rigorous dataset audits, responsible model use, and exploration of frameworks for external auditing and bias mitigation.

IV. PROPOSED SOLUTION

Novel Contribution: Multi-Scale Deblurring with Adaptive Attention Mechanism Overview: To address the limitations identified, we propose a novel approach that incorporates a multi-scale deblurring architecture combined with an adaptive attention mechanism. This approach aims to enhance the model's ability to handle various types of motion blur and improve the overall quality of the deblurred images.

A. Key Features:

- **Multi-Scale Architecture:** By processing images at multiple scales, the model can capture both fine details and broader contextual information. This allows for better handling of different types of blur, including complex and non-linear motion blurs.
- **Adaptive Attention Mechanism:** Integrating an attention mechanism that adapts based on the content of the image can help the model focus on critical areas that require more detail restoration. This can improve the quality of deblurring in regions with significant motion blur while preserving the overall structure of the image.
- **Enhanced Training Dataset:** We propose to augment the training dataset with synthetic images generated from various motion trajectories, including those that mimic real-world scenarios. This will improve the model's robustness and generalization capabilities.